



Introduction to Finance

Fundamentals of Risk in Capital Budgeting

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1 A positive NPV is not enough

Modern project evaluation doesn't just ask if the NPV (Net Present Value) is positive; it asks what the probability is of it being positive and which variables explain that outcome.

The NPV of a project is a random variable whose value depends on the future realization of cash flows. Evaluating a project, therefore, involves analyzing not only its expected value but also the uncertainty associated with that value.

During both the preparation and evaluation stages of a project, various assumptions are required regarding:

- Demand
- Input prices
- Competition
- Interest rates
- Exchange rates
- Inflation
- Cost overruns
- Delays beyond the original schedule
- Technological obsolescence
- Tax changes
- Changes in environmental regulations

2 How to incorporate these risks into project evaluation?

Using a higher discount rate: The higher the risk, the higher the expected (required) return. But how do you determine the specific discount rate to use?

Sensitivity Analysis and Break-even Point: What happens to the NPV if sales drop by 10%? What is the minimum sales volume required for the NPV to equal zero?

Scenario Analysis: Create three or more distinct "worlds": Optimistic, Pessimistic, and Expected. Calculate the NPV for each.

Monte Carlo Simulation: Use software to run thousands of scenarios by combining all random variables to determine the actual probability that the NPV will be greater than zero.

3 Sensitivity Analysis and Break-even Point

3.1 What It Is

Sensitivity analysis is a method that involves modifying the value of one variable at a time—for example, price, quantity demanded, or unit variable cost—while keeping all others constant. By doing this, it does not consider the simultaneous interaction between variables, but rather isolates the individual effect of each one on the project’s profitability.

In financial terms, sensitivity analysis corresponds to a *what-if* exercise: it allows for the evaluation of the marginal impact of changes in a single variable, but it does not provide information regarding the probability of those changes occurring.

3.2 What It Answers

Sensitivity analysis seeks to answer questions such as:

- How sensitive is the NPV to changes in the project’s key variables (e.g., prices, sales volume, fixed costs, etc.)?

Break-even analysis is a component of a project’s sensitivity analysis and seeks to answer:

- What is the maximum or minimum value a variable can take for the NPV to equal zero?

In other words, the break-even point allows you to determine the project’s “waterline”—the threshold at which it begins to become profitable.

3.3 Strengths

- It allows for measuring the project’s tolerance toward changes in key variables.
- It facilitates the identification of economic viability thresholds (critical points).
- It is a simple and intuitive method, useful as an initial project diagnosis.

3.4 Limitations

- By modifying only one variable at a time, it ignores joint effects and interactions between variables.
- It does not provide probabilities: it doesn't indicate how likely it is for a variable to reach its critical value.
- The selection of variables to analyze depends on the evaluator's judgment, making the method intrinsically subjective.
- It does not measure risk; rather, it measures how much a variable can change before the project is no longer acceptable.

Example 1:

- Initial investment ($t=0$): $I_0 = 12.000$
- Horizon: 3 years
- Discount rate: $r = 10\%$
- Tax rate: $t=25\%$
- Quantity: $Q = 500$ units/year
- Price: $P = 40$ \$/unit
- Variable cost: $VC = 20$ \$/unit
- Fixed cost: $FC = 2.000$ \$/year
- Depreciation = 4000

(a) Calculate the cash flow

For $t = 1, 2, 3$ the cash flow is:

$$CF_t = [(20000 - 20 \times 500 - 2000) - 4000] \times (1 - 0.25) + 4000 = 7000$$

(b) Calculate the price (P) where $NPV = 0$

For any value of P , the cash flow is:

$$CF_t = [(500P - 20 \times 500 - 2000) - 4000] \times (1 - 0.25) + 4000 = 375P - 8000$$

$$NPV = -I_0 + \frac{FC}{r} \times \left[1 - \frac{1}{(1+r)^t} \right] = 0$$

Plugging the values, we have

$$-12000 + \frac{375P - 8000}{0.1} \times \left[1 - \frac{1}{1.1^3}\right] = 0$$

$$-12000 + (375P - 8000) \times 2.4869 = 0$$

Solving for P , we have $P = 34.2$

- (c) If $P = \$40$ per unit, calculate the minimum Q where $NPV = 0$. For $t = 1, 2, 3$, the cash flow is equal to:

$$CF_t = [(40Q - 20Q - 2000) - 4000] \times (1 - 0.25) + 4000 = 15Q - 500$$

Substituting the values,

$$-12000 + \frac{15Q - 500}{0.1} \times \left[1 - \frac{1}{1.1^3}\right] = 0$$

$$-12000 + (15Q - 500) \times 2.4869 = 0$$

Solving for Q , we have $Q = 355.02$

4 Scenario Analysis

4.1 What It Is

Scenario analysis is a method that allows for the evaluation of how a project's profitability changes in response to joint variations in a set of key variables, while keeping the model's structure fixed.

Each scenario is constructed from a consistent combination of values for relevant variables—such as prices, quantities, or costs—that reflect a plausible narrative of how the project's environment might evolve. In practice, it is common to define pessimistic, base, and optimistic scenarios, or scenarios of high and low demand, among others.

A scenario does not intend to predict the future, but rather to explore how different plausible stories translate into different economic outcomes.

4.2 What It Answers

Scenario analysis helps answer questions such as:

- How does the project's NPV change if several variables simultaneously behave in an adverse or favorable manner?
- What is the range of possible economic outcomes under different coherent assumptions?
- How sensitive is the project's profitability to joint changes in key variables?

4.3 Strengths

- It allows for the analysis of the joint effects of several variables, unlike one-dimensional sensitivity analysis.
- It is an intuitive and communicatively powerful tool, especially useful for discussing results with decision-makers.
- It facilitates the exploration of extreme but plausible scenarios, helping to understand the project's exposure.

4.4 Limitations

- There is no objective criterion for defining how many scenarios to analyze or which ones are the most relevant.
- It allows for the visualization of possible ranges of profitability, but it does not quantify the probability of the project's success or failure.

Example 2:

	Year 1		Year 2	
	Good	Bad	Good	Bad
Probability	0,9	0,1	0,9	0,1
Revenues	200	200	250	250
Expenses	100	175	125	200
Net cash flow	100	25	125	50

(a) Calculate the expected CF for years 1 and 2

$$ECF_1 = 100 \times 0.9 + 25 \times 0.1 = 92.5$$

$$ECF_2 = 125 \times 0.9 + 50 \times 0.1 = 117.5$$

(b) If the discount rate is 12%, calculate the NPV

$$NPV = \frac{92.5}{1 + 0.12} + \frac{117.5}{(1 + 0.12)^2} = 176.3$$

5 Monte Carlo simulation

5.1 What It Is

Monte Carlo simulation is a stochastic model that allows for project risk analysis by generating a large number of simulations. In this approach, each relevant cash flow variable is assigned a probability distribution that represents its future behavior.

Based on these distributions, the model generates thousands of random yet coherent scenarios in which variables take on values compatible with the defined assumptions. Each scenario produces a profitability outcome, usually the NPV.

In this way, Monte Carlo simulation allows for an approximation of the NPV distribution and, therefore, makes it possible to report the arithmetic mean, standard deviation or variance, distribution percentiles, the range in which the NPV is negative, etc.

5.2 What it Answers

Monte Carlo simulation answers questions that cannot be addressed by deterministic methods, such as:

- What is the probability that the NPV will be negative?
- What is the expected NPV of the project and how dispersed are its potential outcomes?
- Which variables actually explain the project risk and to what magnitude?

5.3 Strengths

Unlike other methods, simulation does not provide a single profitability value, but rather a complete distribution of the NPV, which allows for:

- Estimating probabilities of success and failure.
- Measuring the dispersion and skewness of the results.
- Identifying critical variables through correlation analysis with profitability.

Furthermore, by relying on explicit probabilistic assumptions, simulation reduces evaluator subjectivity and provides a more solid foundation for decision-making.

5.4 Limitations

The quality and usefulness of the results depend directly on:

- The correct modeling of cash flows and the relationships between variables.
- The quality and realism of the assigned probabilistic assumptions.

Monte Carlo simulation does not eliminate risk; rather, it improves the information available for deciding under risk, strengthening the economic analysis of the project.

Example 3:

Assume the project cash flows are normally distributed based on the following information:

- CF_1 has a mean of \$1,800 and a standard deviation of \$180.
 - CF_2 has a mean of \$500 and a standard deviation of \$50.
 - Initial Investment = \$2,000 (fixed).
 - Discount Rate = 10% (fixed).
- (a) In a deterministic model, calculate the NPV assuming the cash flows are equal to their means and the standard deviation is zero.

$$NPV = -2000 + \frac{1800}{(1 + 0.1)} + \frac{500}{(1 + 0.1)^2} = 50$$

Answer (b)-(e), assuming that the cash flows follow a normal distribution:

- (b) Determine the probability distribution of the NPV. What is the expected NPV? What is the standard deviation (volatility)?

- (c) What is the probability that the NPV is less than zero?
- (d) Which cash flows primarily drive the project risk, and what is their magnitude of impact?
- (e) How would you model an increase in the uncertainty of CF_2 ? how does this change affect the expected NPV and the volatility??

This involves increasing its dispersion around the mean; for example, increasing the standard deviation from \$50 to \$100.

- (f) What are the advantages of utilizing a triangular distribution rather than a normal distribution when modeling project cash flows?
 - *Boundary definition:* A triangular distribution allows you to set a strict minimum and maximum cash flow. A normal distribution is theoretically infinite, which could result in a simulation generating impossible cash flow values (e.g., extreme losses that are physically or contractually impossible).
 - *Skewness and asymmetry:* Normal distributions are symmetrical around the mean. Triangular distributions can be skewed, allowing you to model cash flows where the “downside” risk is larger than the “upside” potential (or vice versa).
 - *Intuitive estimation:* It is often more practical for a manager to estimate the “minimum” “maximum” and “most likely (mode)” cash flow values than to accurately estimate a standard deviation for a normal curve.
- (g) Using the triangular distribution, how would you model cash flows that are positively skewed? Provide an example of company that might fit this model.

A positive skew (or right-skewed distribution) occurs when the “tail” on the right side is longer than the left. In financial modeling, this means that while the most likely outcome is relatively low, there is a significant “upside” potential. To achieve this place the *mode* much closer to the *minimum* than to the *maximum*.

A biotech startup in the early stages of drug development is a classic example of a company that fits a positively skewed model.

- Most Likely Outcome (Mode): The drug fails to pass clinical trials or achieves only modest sales. Cash flows remain low.
- Minimum (Worst Case): Total failure; the project is scrapped, and cash flow is zero or slightly negative due to shutdown costs.
- Maximum (Best Case): The drug becomes a “blockbuster” success, capturing a massive global market. This creates a “long tail” of extreme profitability.

Other examples of positive skewness include oil exploration firms and venture capital funds. On the other hand, companies such as water and electricity utilities, airlines, and insurance firms typically have negatively skewed cash flows.

- (h) Construct a numerical example by assuming that the project's cash flows follow a triangular distribution. Determine the expected NPV, the volatility, and the 10th and 90th percentiles of the NPV distribution.

6 Decision trees

This technique allows for the representation and analysis of a series of sequential future decisions over time. The model uses a graphical structure where:

- *Squares* represent decision points (combinations of choices).
- *Circles* represent random events that influence outcomes.

6.1 What They Answer

Analysis through *decision trees* seeks to determine the optimal course of action when facing multiple options and future scenarios, allowing for:

- Estimating an expected value for the NPV, profit, or any other relevant financial indicator.
- Evaluating whether it is convenient to execute decision A before B or C, depending on intermediate results.

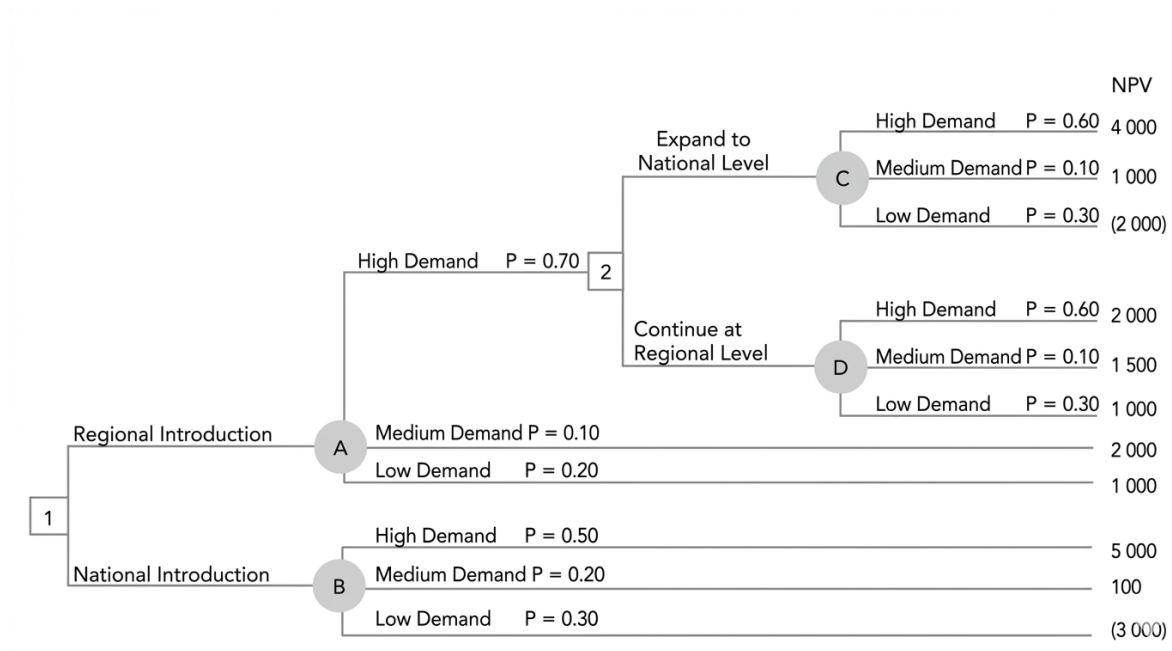
6.2 Strengths

- It allows for the graphical visualization of all possible project paths and their consequences.
- It is ideal for projects that require staged decision-making.

Example 4:

A company is evaluating the launch of a new product, either nationally or regionally. If it decides to launch the product in one region, it is possible to expand nationally later if the initial results recommend doing so.

The decision tree is represented below:



What decision would you make?

To determine the best combination of decisions, you begin at the end of the branches (the outcomes on the right) and move backward toward the origin (the decision on the left).

$$\text{Node C: } 0.6 \times 4000 + 0.1 \times 1000 + 0.3 \times -2000 = 1900$$

$$\text{Node D: } 0.6 \times 2000 + 0.1 \times 1500 + 0.3 \times 1000 = 1650$$

At decision Node 2, the optimal strategy is to *Expand to National Level*, as it maximizes the Expected NPV.

$$\text{Node A: } 0.7 \times 1900 + 0.1 \times 2000 + 0.2 \times 1000 = 1730$$

$$\text{Node B: } 0.5 \times 5000 + 0.2 \times 100 + 0.3 \times -3000 = 1620$$

At decision Node 1, the optimal strategy is *Regional Introduction*, as it maximizes the Expected NPV.

Therefore, the first choice is a regional introduction, which is then expanded to the national level.

6.3 Limitations

- As the number of sequential decisions and possible random events increases, the tree becomes extremely complex and difficult to manage visually.
- Its usefulness depends on the quality of the probabilities assigned to each branch, which are often subjective or based on historical data that may not repeat.
- Although it considers different paths, each branch usually ends in a single point value for NPV, ignoring the full distribution of results that a Monte Carlo Simulation provides.
- It does not always effectively capture the interactions between variables that affect different branches of the tree simultaneously.